SAP MM • S/4HANA 2023

Open PO Monitoring & Aging Report

Custom ABAP Report with Traffic Light Aging (T-code: ZPO_AGING)

Project – Training System

System	SAP S/4HANA 2023
Module	SAP MM / Purchasing
Development	Custom ABAP – SE38 + SE93
T-code	`ZPO_AGING`
Environment	SAP Training System (Demo)

Note: Developed in SAP Training System for demonstration and portfolio purposes.

1. Project Overview

Designed and implemented a custom SAP MM ABAP report to monitor open Purchase Orders and track the aging of outstanding PO line items. The solution (program ZOPEN_PO_AGING) provides real-time visibility into overdue and pending PO lines across plants and purchasing organizations, with traffic light indicators for instant prioritization.

The report was built using ABAP with ALV Grid output and deployed via a custom transaction code ZPO_AGING through SE93, making it directly executable by procurement and finance teams.

2. Business Requirement

Procurement and finance teams lacked a consolidated view of open Purchase Orders and their aging status. Standard SAP transactions provided list views but required manual filtering with no aging classification — making it difficult to identify overdue deliveries, follow up with vendors, and manage procurement commitments.

Dimension	Detail
Problem	No single report for open PO aging; manual date calculations required
Affected Teams	Procurement and finance
Current Workaround	Manual cross-referencing across ME2M, ME2N, and ME2L
Impact	Missed delivery deadlines, untracked procurement commitments, delayed vendor follow-up

3. Objectives

- Monitor all open PO line items across plants and purchasing organizations
- Classify PO lines by aging status using traffic light indicators (Green / Yellow / Red)
- Identify overdue deliveries based on scheduled delivery date vs. system date
- Support vendor follow-up and procurement escalation decisions
- Provide a reusable, executable report via custom transaction code

4. Scope

	Items
In Scope	Open PO lines with pending delivery quantity, aging classification by delivery date, vendor details, plant and company code visibility, user execution via T-code
Out of Scope	Automatic vendor communication, GR posting from report, integration with Ariba or external procurement systems

5. Solution Overview

Custom ABAP report ZOPEN_PO_AGING built by joining four database tables to retrieve all open PO lines with delivery schedule and vendor details:

Table	Name	Purpose
EKKO	PO Header	Purchase Order header data including vendor and company code
EKPO	PO Item	Purchase Order item data including quantities and material info
EKET	Delivery Schedule	Scheduled delivery dates and GR quantities per PO line
LFA1	Vendor Master	Vendor name and details for display in output

Output rendered via REUSE_ALV_GRID_DISPLAY with zebra striping, column optimization, traffic light column, and Excel export capability. Custom transaction ZPO_AGING created in SE93 for direct end-user execution.

6. Functional Logic

Step	Logic
1. Selection Input	User enters Company Code, Plant, and optional PO Number range on the selection screen
2. Data Fetch	Report joins EKKO/EKPO with LFA1 and EKET; deleted lines (LOEKZ) and fully received lines (ELIKZ) are excluded
3. Open Qty Calculation	Open qty = PO Quantity − GR Quantity; lines with open qty ≤ 0 are removed from output
4. Aging Calculation	Age in days = SY-DATUM − EINDT (scheduled delivery date)
5. Traffic Light	Green ≤ 30 days; Yellow 31–90 days; Red 90+ days — drives visual status column in ALV
6. ALV Output	Displayed in ALV Grid with sort, filter, column optimization, and Excel export capability

7. Aging & Traffic Light Reference

Status	Aging Threshold	Business Meaning
● Green	0 – 30 days	Delivery within acceptable window; no action required
● Yellow	31 – 90 days	Approaching overdue; monitor and initiate vendor follow-up
● Red	90+ days	Overdue; escalate to procurement management immediately

8. Report Fields

8.1 Selection Screen Fields

Field	Description
Company Code (BUKRS)	Filter by company code
Plant (WERKS)	Filter by plant
Purchase Order Number	Optional range restriction on PO number (S_EBELN)

8.2 Output Fields

Field	Source / Description
Status (Traffic Light)	Calculated – aging-based indicator (Green/Yellow/Red)
PO Number	EKKO-EBELN
Item	EKPO-EBELP
Vendor	LFA1-NAME1 – vendor name
Material	EKPO-MATNR
Description	EKPO-TXZ01 – short text
PO Quantity	EKPO-MENGE
GR Quantity	EKET-WEMNG – goods receipt quantity
Open Quantity	Calculated: PO Qty − GR Qty
Delivery Date	EKET-EINDT – scheduled delivery date
Age (in Days)	Calculated: SY-DATUM − EINDT

9. Business Benefits

Benefit	Detail
Consolidated View	Replaces manual cross-referencing of ME2M, ME2N, and ME2L with a single executable report
Instant Prioritization	Traffic light indicators allow procurement teams to act without manual date calculations
Overdue Visibility	Reduces risk of missed deliveries impacting production or warehouse replenishment
Excel Export	ALV Grid supports direct Excel export for reporting and escalation workflows
ABAP Showcase	Demonstrates ABAP development capability applied to a real procurement monitoring use case

10. Technical Artifacts

Artifact	Value
Program Name	ZOPEN_PO_AGING
Transaction Code	ZPO_AGING
Tables Used	EKKO · EKPO · EKET · LFA1
Output Type	ALV Grid (REUSE_ALV_GRID_DISPLAY)
Created Via	SE38 + SE93
Source Code	268 lines – active and executable in SAP S/4HANA 2023 Training System
Authorization	Standard user access in training system

11. ABAP Code Walkthrough — ZOPEN_PO_AGING

> *The report is structured into logical sections following standard ABAP conventions. Key sections are documented below. Full source code (268 lines) available on request — program is active and executable in SAP S/4HANA 2023 Training System.*

11.1 Report Declaration & Table References

Declares the program, activates SLIS type pools for ALV, and references the database tables used in the SELECT statement.

```
REPORT ZOPEN_PO_AGING.
TYPE-POOLS: slis.
TABLES: ekko, ekpo, eket, lfa1.
```

11.2 Selection Screen

Allows the user to filter by Company Code (BUKRS), Plant (WERKS), and optionally by PO Number range. Keeps the selection screen simple and user-friendly.

```
PARAMETERS: p_bukrs TYPE ekko-bukrs,
            p_werks TYPE ekpo-werks.
SELECT-OPTIONS: s_ebeln FOR ekko-ebeln.
```

11.3 Aging Calculation & Traffic Light Logic

Open quantity is calculated as PO quantity minus GR quantity. Aging in days is derived using SY-DATUM minus scheduled delivery date (EINDT). Traffic light values drive the ALV status column.

```
DATA lv_age TYPE i.

LOOP AT gt_po INTO gs_po.

  "Open Quantity
  gs_po-openq = gs_po-menge - gs_po-wemnq.
  IF gs_po-openq < 0.
    gs_po-openq = 0.
  ENDIF.

  "Age Calculation
  lv_age    = sy-datum - gs_po-eindt.
  gs_po-age = lv_age.

  MODIFY gt_po FROM gs_po INDEX sy-tabix
               TRANSPORTING openq age light.

ENDLOOP.

DELETE gt_po WHERE openq <= 0.

"Traffic Light Logic
  IF lv_age <= 30.
    gs_po-light = '3'.    "Green
  ELSEIF lv_age <= 90.
    gs_po-light = '2'.    "Yellow
  ELSE.
    gs_po-light = '1'.    "Red
  ENDIF.
```

11.4 Data Fetch

Joins EKKO, EKPO, LFA1, and EKET in a single SELECT to retrieve all open PO lines matching the selection criteria. Filters out deleted lines (loekz = '') and already fully-received lines (elikz = '').

```
CALL FUNCTION 'REUSE_ALV_GRID_DISPLAY'
  SELECT a~ebeln b~ebelp a~lifnr c~name1
       b~matnr b~txz01 b~menge d~wemng d~eindt
  INTO CORRESPONDING FIELDS OF TABLE gt_po
  FROM ekko AS a
  INNER JOIN ekpo AS b ON a~ebeln = b~ebeln
  INNER JOIN lfa1 AS c ON a~lifnr = c~lifnr
  INNER JOIN eket AS d ON b~ebeln = d~ebeln
                      AND b~ebelp = d~ebelp
  WHERE a~bukrs   = p_bukrs
    AND b~werks   = p_werks
    AND a~ebeln  IN s_ebeln
    AND b~loekz   = ''
    AND b~elikz   = ''.
```

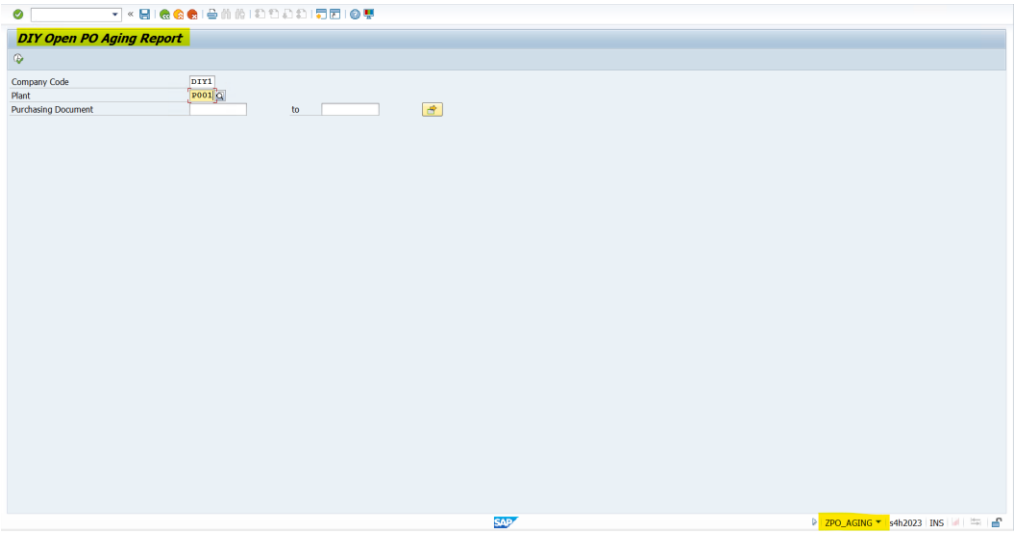
11.5 ALV Grid Display

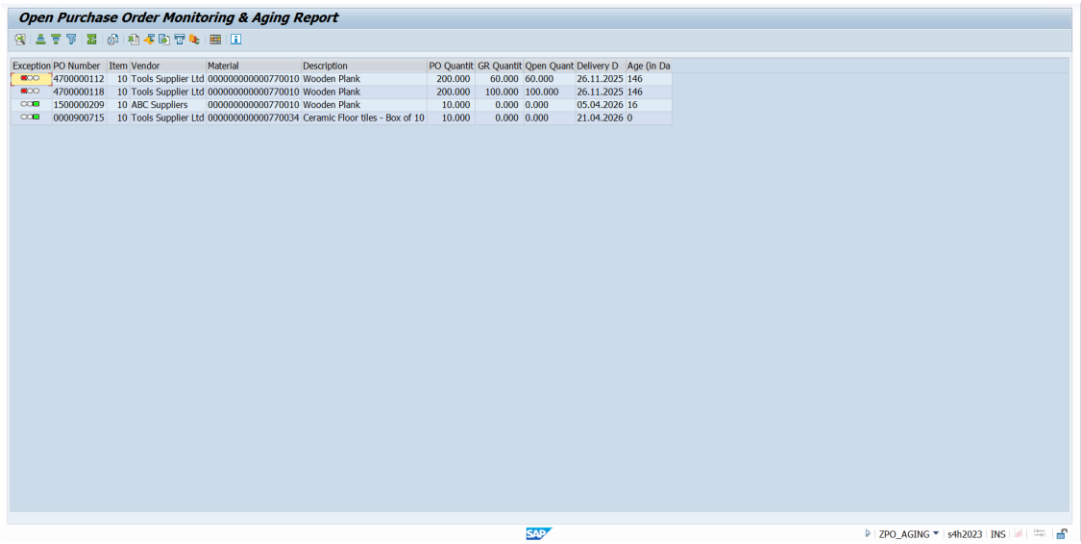
Calls REUSE_ALV_GRID_DISPLAY to render the output with the configured layout and field catalog. Zebra striping, column optimisation, and the traffic light column are enabled through layout settings.

```
CALL FUNCTION 'REUSE_ALV_GRID_DISPLAY'
  EXPORTING
    I_CALLBACK_PROGRAM      = 'sy-repid'
    I_CALLBACK_USER_COMMAND = 'USER_COMMAND'
    IS_LAYOUT               = gs_layout
    IT_FIELDCAT             = gt_fieldcat
  TABLES
    t_outtab                = gt_po.  PERFORM add_field USING 'MATNR' 'Material'.
```

12. Testing & Validation

- Execute report for Company Code and Plant with no PO number restriction
- Verify open PO lines appear with correct open quantity (PO qty minus GR qty)
- Confirm fully received lines are excluded from output
- Validate aging days calculated correctly against system date
- Check traffic light colours reflect correct aging thresholds (Green ≤30, Yellow ≤90, Red 90+)
- Export ALV output to Excel and verify all fields populate correctly
- Execute via custom T-code ZPO_AGING and confirm report launches directly





✓ Validation Passed – All aging thresholds, open quantity calculations, and traffic light indicators validated successfully.

Note: Sample data shown for demo purposes. Program is active and executable in SAP S/4HANA 2023 Training System. Full source code (268 lines) available on request.